

PHASE2: SIEM ANALYSIS



جامعة الملك فهد للبترول والمعادن
King Fahd University of Petroleum & Minerals

OVERVIEW



Abstract

In this report, we focused on the configuration of Splunk, log collection from the victim and attacker environments, and the use of Splunk's visualization and analysis capabilities to get analysis into the SSH attack conducted in Phase 1.



Requirements

- **Need #1: SIEM Setup:** Integrate logs from both the victim and attacker environments into a SIEM tool (e.g., Splunk).
- **Need #2: Log Visualization:** Visualize and analyze attack logs to understand attack patterns and behaviors.
- **Need #3: Attack Comparison:** Compare data from the victim and attacker environments.



Deliverables

- **Need #1:** Screenshot of SIEM integration (logs from both environments).
- **Need #2:** Screenshot of attack visualization and analysis.

TASK 1: SPLUNK INSTALL (IN ATTACKER VM)



1A. Splunk download

First i download the splunk server

```
(kali㉿kali)-[~]  
$ wget -O splunk-9.3.2-d8bb32809498-linux-2.6-amd64.deb https://download.splunk.com/products/splunk/releases/9.3.2/linux/splunk-9.3.2-d8bb32809498-linux-2.6-amd64.deb
```



1B. Splunk install

```
(kali㉿kali)-[~]  
$ sudo dpkg -i splunk-9.3.2-d8bb32809498-linux-2.6-amd64.deb
```



1C. start splunk

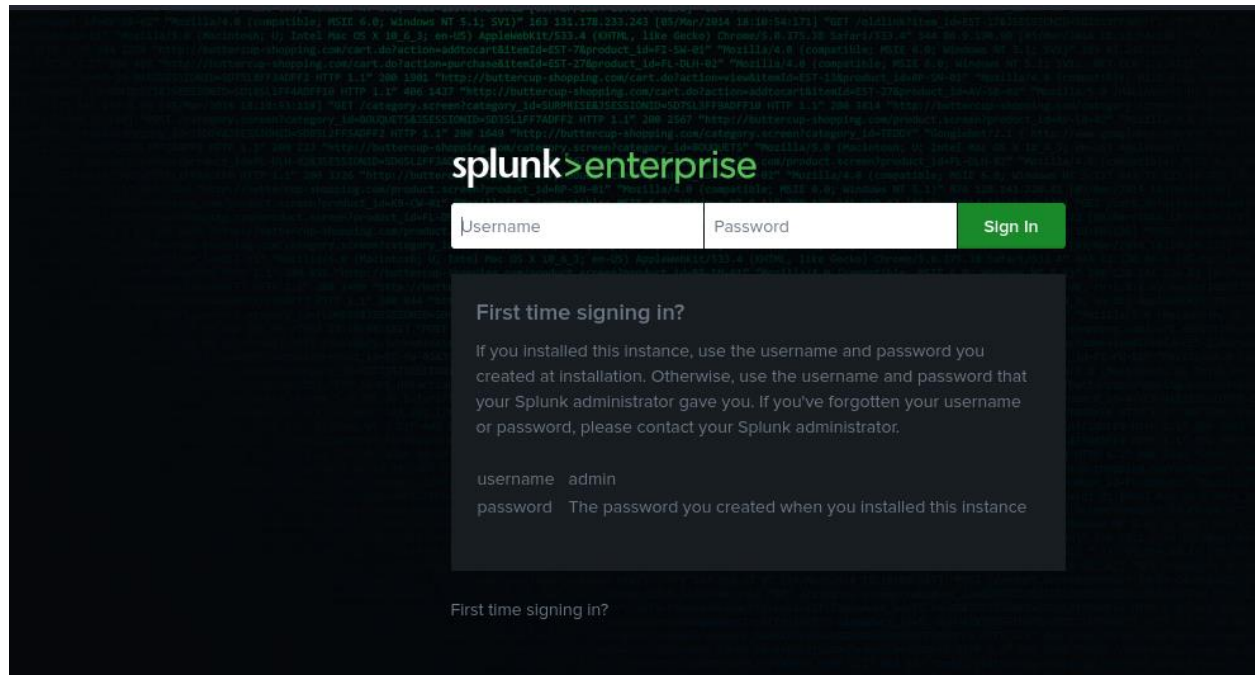
```
kali@kali: ~  
File Actions Edit View Help  
Checking conf files for problems...  
Done  
Checking default conf files for edits...  
Validating installed files against hashes from '/opt/splunk/splunk-9.3.2-d8bb32809498-linux-2.6-x86_64-manifest'  
All installed files intact.  
Done  
All preliminary checks passed.  
  
Starting splunk server daemon (splunkd)...  
PYTHONHTTPSVERIFY is set to 0 in splunk-launch.conf disabling certificate validation for the httplib and urllib libraries shipped with the embedded Python interpreter; must be set to "1" for increased security  
Done  
  
Waiting for web server at http://127.0.0.1:8000 to be available..... Done  
  
If you get stuck, we're here to help.  
Look for answers here: http://docs.splunk.com  
  
The Splunk web interface is at http://kali:8000
```



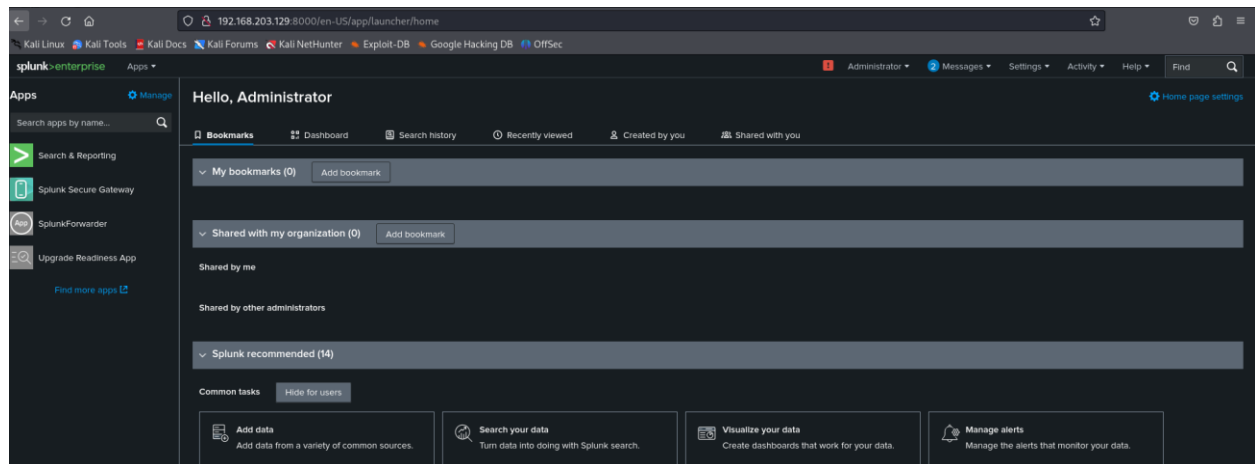
1D. enter splunk server website using the attacker ip :

Attacker ip here is : 192.168.203.129

I sign in using my username and password :



Then it shows me the main page :



TASK 2: SPLUNK FORWARDER INSTALL (FROM VICTIM VM)



2A. download splunk forwarder and install it

Download and install

```
vagrant@metasploitable3-ub1404:~$ uget -O splunkforwarder-9.4.1-e3bdab203ac8-linux-amd64.deb "https://download.splunk.com/products/universalforwarder/releases/9.4.1/linux/splunkforwarder-9.4.1-e3bdab203ac8-linux-amd64.deb"
--2025-04-29 21:35:56-- https://download.splunk.com/products/universalforwarder/releases/9.4.1/linux/splunkforwarder-9.4.1-e3bdab203ac8-linux-amd64.deb
Resolving download.splunk.com (download.splunk.com)... 108.159.236.91, 108.159.236.108, 108.159.236.116, ...
Connecting to download.splunk.com (download.splunk.com):108.159.236.91:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 99029222 (94M) [binary/octet-stream]
Saving to: 'splunkforwarder-9.4.1-e3bdab203ac8-linux-amd64.deb'

100%[=====>] 99,029,222  1.63MB/s   in 57s

2025-04-29 21:36:54 (1.66 MB/s) - 'splunkforwarder-9.4.1-e3bdab203ac8-linux-amd64.deb' saved [99029222/99029222]

vagrant@metasploitable3-ub1404:~$ sudo dpkg -i splunkforwarder-9.4.1-e3bdab203ac8-linux-amd64.deb
Selecting previously unselected package splunkforwarder.
(Reading database ... 127930 files and directories currently installed.)
Preparing to unpack splunkforwarder-9.4.1-e3bdab203ac8-linux-amd64.deb ...
no need to run the pre-install check
Unpacking splunkforwarder (9.4.1) ...
Setting up splunkforwarder (9.4.1) ...
find: '/opt/splunkforwarder/lib/python3.7/site-packages': No such file or directory
find: '/opt/splunkforwarder/lib/python3.9/site-packages': No such file or directory
complete
vagrant@metasploitable3-ub1404:~$ sudo apt --fix-broken install
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  amd64-microcode linux-modules-extra-3.13.0-170-generic
Use 'apt-get autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 5 not upgraded.
vagrant@metasploitable3-ub1404:~$ _
```



2B. run splunk forwarder

```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk start --accept-license
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
The splunk daemon (splunkd) is already running.
```

TASK 3: SPLUNK SERVER SETUP (IN ATTACKER VM "KALI")



3A. Configure inputs.conf file to let splunk listen on port 9997 for the incoming data :

```
kali@kali: ~  
File Actions Edit View Help  
GNU nano 8.4 /opt/splunk/etc/system/local/inputs.conf  
[default]  
host = $decideOnStartup  
  
[tcp://9997]  
connection_host = none  
sourcetype = ssh  
[splunktcp://9997]  
disabled = 0
```

TASK 4: SPLUNK FORWARDER LOG SETUP (VICTIM VM "METASPLOIT3")



4A. we edit output.conf file to tell the forwarder where to send data

We configure the outputs.conf file on victim vm to let Forwarder send data to the Splunk server:

```
GNU nano 2.2.6 File: /opt/splunkforwarder/etc/system/local/outputs.conf  
  
[tcpout]  
defaultGroup = default-autolb-group  
  
[tcpout:default-autolb-group]  
server = 192.168.203.129:9997  
  
[tcpout-server://192.168.203.129:9997]
```



4B. Connect the forwarder to the splunk server:

```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk add forward-server 192.168.203.129:9997
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Splunk username: mo.alzaher1@gmail.com
Password:
Login failed
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk add forward-server 192.168.203.129:9997
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Splunk username: vagrant
Password:
Added forwarding to: 192.168.203.129:9997.
vagrant@metasploitable3-ub1404:~$
```



4C. Add log files to the splunk server:

```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Cannot create another input with the name "/var/log/auth.log", one already exists.
vagrant@metasploitable3-ub1404:~$
```



4D. Check forwarder status:

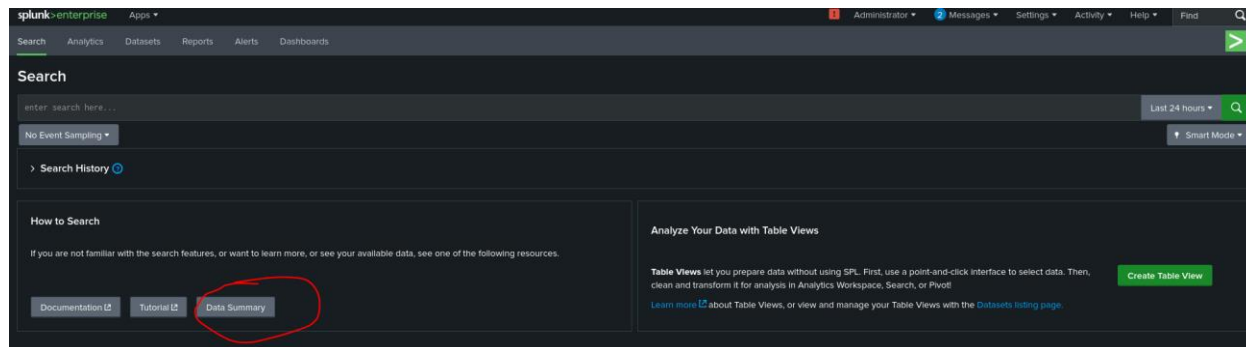
We can see that our forwarder is actively forwarding logs to server at 192.168.203.129:9997

```
vagrant@metasploitable3-ub1404:~$ sudo /opt/splunkforwarder/bin/splunk list forward-server
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"
Active forwards:
    192.168.203.129:9997
Configured but inactive forwards:
    None
```

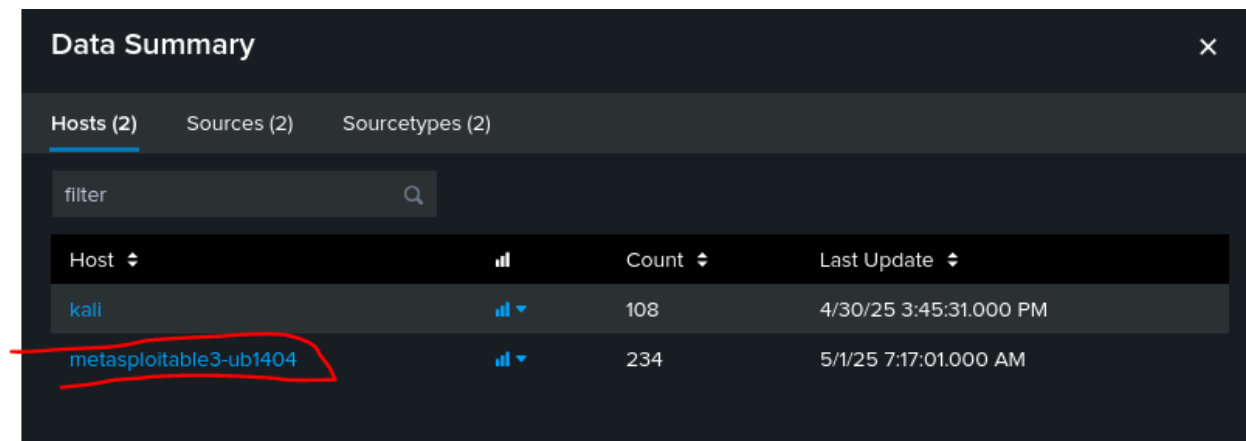
TASK 5: SHOW DATA RECEIVED IN SPLUNK SERVER



5A. first we go to the data summary in splunk server:



5B. then we choose our victim vm metasploitable3-ub1404



5C. then we can see the forwarded data that comes from splunk forwarder to the splunk server:

List	Format	50 Per Page	< Prev	1	2	3	4	5	Next >
i	Time	Event							
>	4/30/25 3:58:57.000 PM	Apr 30 19:58:57 metasploitable3-ub1404 sudo: pam_unix(sudo:session): session closed for user root host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog							
>	4/30/25 3:58:57.000 PM	Apr 30 19:58:57 metasploitable3-ub1404 sudo: pam_unix(sudo:session): session opened for user root by vagrant(uid=0) host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog							
>	4/30/25 3:58:57.000 PM	Apr 30 19:58:57 metasploitable3-ub1404 sudo: vagrant : TTY=pts/0 ; PWD=/home/vagrant ; USER=root ; COMMAND=/opt/splunkforwarder/bin/splunk add monitor /var/log/auth.log host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog							
>	4/30/25 3:58:06.000 PM	Apr 30 19:58:06 metasploitable3-ub1404 sshd[3213]: Connection closed by 192.168.203.129 [preauth] host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog							
>	4/30/25 3:58:06.000 PM	Apr 30 19:58:06 metasploitable3-ub1404 sshd[3213]: Failed password for invalid user vagrant # Default username for Metasploitable3 from 192.168.203.129 port 41051 ssh2 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog							
>	4/30/25 3:58:04.000 PM	Apr 30 19:58:04 metasploitable3-ub1404 sshd[3213]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh ruser= rhost=192.168.203.129 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog							
>	4/30/25 3:58:04.000 PM	Apr 30 19:58:04 metasploitable3-ub1404 sshd[3213]: pam_unix(sshd:auth): check pass; user unknown host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog							
>	4/30/25 3:58:04.000 PM	Apr 30 19:58:04 metasploitable3-ub1404 sshd[3213]: input_userauth_request: invalid user vagrant # Default username for Metasploitable3 [preauth] host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog							
>	4/30/25 3:58:04.000 PM	Apr 30 19:58:04 metasploitable3-ub1404 sshd[3213]: Invalid user vagrant # Default username for Metasploitable3 from 192.168.203.129 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog							
>	4/30/25	Apr 30 19:57:57 metasploitable3-ub1404 sshd[3209]: Connection closed by 192.168.203.129 [preauth]							



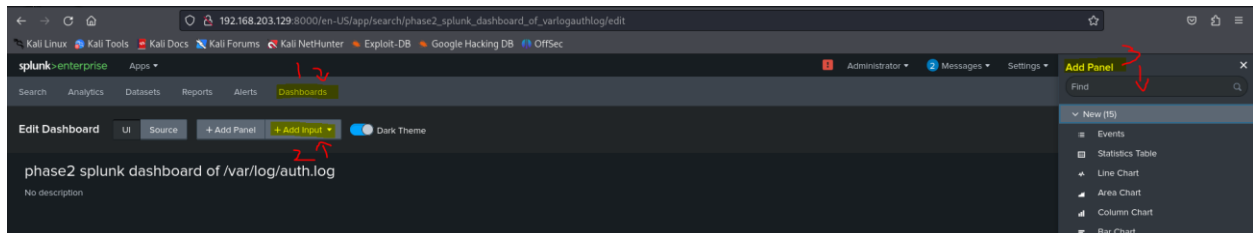
5D. the success and failed password logs:

New Search	Save As	Create Table View	Close
source="/var/log/auth.log" ("failed password" OR "Accepted password") eval Result=if(searchmatch("failed password"), "failed", "Accepted") stats count by Result			Last 24 hours
6 events (4/29/25 3:00:00.000 PM to 4/30/25 3:53:32.000 PM) No Event Sampling	Job		Smart Mode
Events	Patterns	Statistics (2)	Visualization
20 Per Page	Format	Preview	
Result			count
Accepted			2
Failed			4

>	4/30/25 3:44:50.000 PM	Apr 30 19:44:50 metasploitable3-ub1404 sshd[2810]: Accepted password for vagrant from 192.168.203.129 port 42878 ssh2 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog
>	4/30/25 3:58:06.000 PM	Apr 30 19:58:06 metasploitable3-ub1404 sshd[3213]: Failed password for invalid user vagrant # Default username for Metasploitable3 from 192.168.203.129 port 41051 ssh2 host = metasploitable3-ub1404 source = /var/log/auth.log sourcetype = syslog

TASK 6: LOG VISUALIZATION WITH ATTACKS COMPARISON

👉 6A. We added log visualization to analyze attack logs and understand attack patterns. We did it from here: 1- dashboard 2- add input 3- choose the panel that we want.

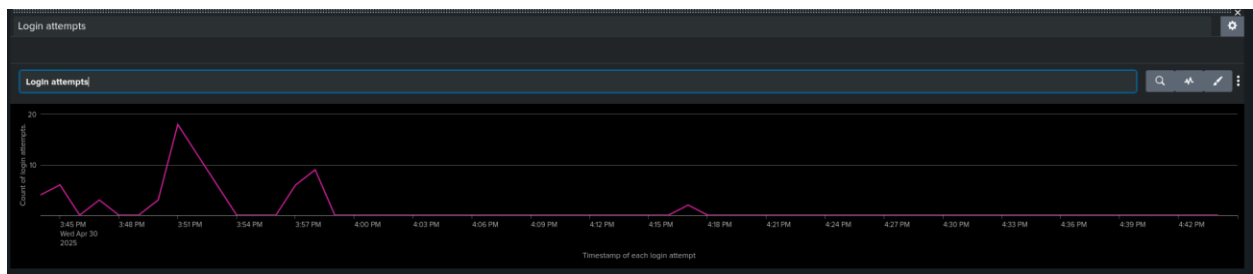


👉 6B. Events table:

phase2 events:										
row #	time	host	index	linecount	source	sourcetype	splunk_server			
May 1 18:27:34	metasploit3-ub1484 sudo: vagrant : TTY=ttty ; PWD=/home/vagrant ; USER=root ; COMMAND=/opt/splunkforwarder	metasploit3-ub1484	main	1	/var/log/auth.log	syslog	kali			
May 1 18:02:03	metasploit3-ub1484 sudo: vagrant : TTY=pts/0 ; PWD=/home/vagrant ; USER=root ; COMMAND=/opt/splunkforwarder	metasploit3-ub1484	main	1	/var/log/auth.log	syslog	kali			
May 1 18:01:49	metasploit3-ub1484 sudo: vagrant : TTY=pts/0 ; PWD=/home/vagrant ; USER=root ; COMMAND=/opt/splunkforwarder	metasploit3-ub1484	main	1	/var/log/auth.log	syslog	kali			
Apr 30 15:58:57	metasploit3-ub1484 sudo: vagrant : TTY=pts/0 ; PWD=/home/vagrant ; USER=root ; COMMAND=/opt/splunkforwarder	metasploit3-ub1484	main	1	/var/log/auth.log	syslog	kali			
Apr 30 15:53:12	metasploit3-ub1484 sudo: vagrant : TTY=pts/0 ; PWD=/home/vagrant ; USER=root ; COMMAND=/opt/splunkforwarder	metasploit3-ub1484	main	1	/var/log/auth.log	syslog	kali			
Apr 30 15:36:34	metasploit3-ub1484 sudo: vagrant : TTY=pts/0 ; PWD=/home/vagrant ; USER=root ; COMMAND=/bin/grep splunk	metasploit3-ub1484	main	1	/var/log/auth.log	syslog	kali			
Apr 30 15:30:43	metasploit3-ub1484 sudo: vagrant : TTY=pts/0 ; PWD=/home/vagrant ; USER=root ; COMMAND=/opt/splunkforwarder	metasploit3-ub1484	main	1	/var/log/auth.log	syslog	kali			
Apr 30 15:30:15	metasploit3-ub1484 sudo: vagrant : TTY=pts/0 ; PWD=/home/vagrant ; USER=root ; COMMAND=/bin/ls -lrt /var/log/auth.log	metasploit3-ub1484	main	1	/var/log/auth.log	syslog	kali			
Apr 30 15:30:06	metasploit3-ub1484 sudo: vagrant : TTY=pts/0 ; PWD=/home/vagrant ; USER=root ; COMMAND=/usr/bin/tail -f	metasploit3-ub1484	main	1	/var/log/auth.log	syslog	kali			
Apr 30 15:28:25	metasploit3-ub1484 sudo: vagrant : TTY=pts/0 ; PWD=/home/vagrant ; USER=root ; COMMAND=/usr/bin/tail -f	metasploit3-ub1484	main	1	/var/log/auth.log	syslog	kali			

👉 6C. login attempts chart

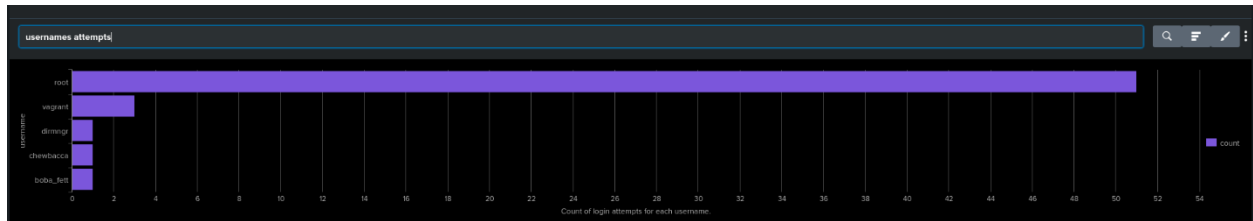
Here we have the login attempts and when they were:



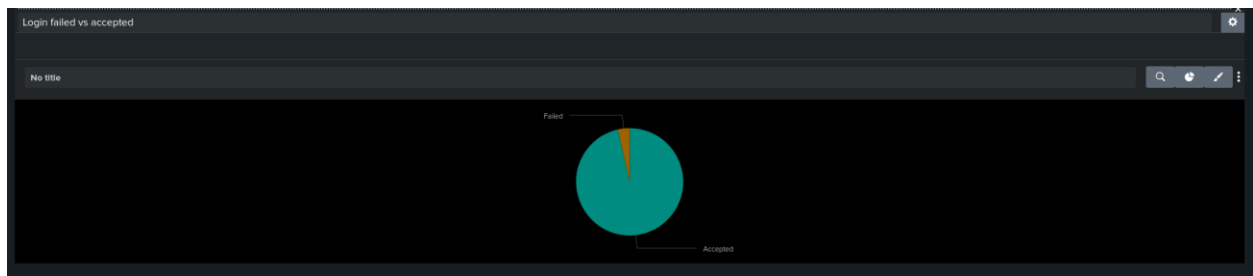


6D. Different username attempts chart:

Here we have different usernames attempts to login



6E. Login failed compared to login success pie chart:



6F. Successful login details table:

Successful Login Details	
_time ↕	
2025-04-30	15:28:21
2025-04-30	15:44:50



6G. here is a failed ssh brute force attack. The username was wrong:

TASK 7: FINAL ANALYSIS

Environment Setup

Two environments were configured (Metasploitable3 and Kali Linux)

1- Victim Environment (where Splunk forwarder is installed here)

- System: Metasploitable3 – Ubuntu 14.04 (ub1404)
- Main tool: Splunk forwarder
- Purpose: to forward logs to the Splunk server
- IP address: 192.168.203.130

2- Attacker Environment (where Splunk server is installed here)

- System: Kali Linux
- Tools Used: Splunk Enterprise
- Purpose: Hosts the Splunk SIEM for log analysis and visualization.
- IP Address: 192.168.203.129

Network Configuration:

- Both VMs were configured to communicate with each other using a Host-Only Adapter to make sure of security and to have isolated communication.
-

Tasks 1,2: Splunk Setup and Log Collection

Splunk Installation:

- We went through the details of the Splunk installation process on both Kali Linux VM and Metasploitable3.

Splunk Forwarder configuration:

- We install Splunk forwarder on the Metasploitable3 VM victim.
- We modified the outputs.conf file configuration and the target Splunk server.

Splunk server configuration:

- We install Splunk Enterprise on the Kali VM attacker.
 - We modified the inputs.conf file to receive incoming logs from 9997 port.
 -
-

Task 3,4: Log collection

Log Collection Method:

- SSH logs from Metasploitable3 were forwarded to the Splunk server on Kali Linux using the Splunk Forwarder.
 - We let the Splunk server monitor logs from Metasploitable3
-

Task 5,6: Visualizations and comparisons

- We visualize the attacks different logs like login attempts and failed compared to successful logins.
 - We use Splunk Query to visualize that
 - Based on these results, we understand how attacks work and how different approaches can lead to different data analysis.
-

Conclusion

In this phase, we successfully analyzed our attacks from phase 1 using splunk enterprise. The log data give us a detailed record of the attacker's actions. This allowed us to understand the attacks times and types in more details. By using splunk, we were able to use its visualization tools to understand the attacks impact and how is its patterns. And it shows us the importance of integrating logging SIEM solutions in the security monitoring.